Report for Project B - Term Deposit Marketing

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1 Executive Summary

Executive Summary The primary objective of this project was to develop a robust and interpretable machine learning system that predicts whether a customer will subscribe to a term deposit (y = yes/no) based on call center data from a European banking institution. The dataset, derived from direct marketing campaigns, contains 40,000 observations with 13 input features and a highly imbalanced target variable (only 7.24% of customers subscribe). The client's baseline requirement was to achieve at least 81% overall accuracy under 5-fold cross-validation.

During the modeling process, it became clear that accuracy alone was misleading and misaligned with the true business goal. A model with high accuracy could simply predict "no" for almost all customers and still exceed 81%, while completely failing to identify the minority class (subscribers). From a business perspective, correctly identifying customers who are likely to subscribe (high recall for y = 1) is far more valuable, as it directly drives higher conversion rates and more efficient allocation of sales resources.

To address this challenge, we selected XGBoost as the best-performing model after extensive comparison. By carefully tuning hyperparameters, applying class weighting, and adjusting the decision threshold, we achieved a balanced trade-off between accuracy and recall. **The final model delivered an average 5-fold cross-validation accuracy of 85.35% (well above the 81% target) while maintaining a recall for the positive class (y = 1) at 91.71%** (with precision = 0.32), significantly outperforming baseline models in capturing potential subscribers. Meanwhile, we also provide substantial methods. To reach a higher recall at 95%, we can sacrifice the accuracy to 81%. Or when the company has no energy to deal with a lot of mistakes, we can also improve the precision to 44%, with a lower recall at 76.52%.

For interpretability, we conducted feature importance analysis using both XGBoost's built-in gain-based importance and SHAP values. This revealed the most influential predictors, with duration (call duration in seconds), previous outcome, month, contact type, and age consistently ranking at the top. These insights enable the bank to prioritize calls with higher expected duration and target specific customer segments more effectively. To further support targeted marketing, we applied K-means clustering exclusively on the successfully subscribed customers (y = 1) and visualized the clusters in two dimensions using t-SNE. The analysis identified four distinct customer segments with clear boundaries. Each segment exhibits unique characteristics in key features (e.g., age, job type, marital status, balance, and campaign contact frequency), providing actionable guidance for personalized outreach strategies.

Overall, this project not only exceeded the accuracy requirement but also delivered a business-aligned, interpretable, and practically useful solution. By focusing on recall

for the positive class and uncovering high-potential customer segments, the resulting system offers the client a clear path to improve call success rates and optimize marketing efficiency.

2 Business Understanding

2.1 Background

The company is a startup specializing in machine learning solutions for the European banking sector, with a focus on applications such as fraud detection, sentiment classification, and customer intention prediction. This project leverages call center data from direct marketing campaigns conducted by a European banking institution. The campaigns involve outbound phone calls to existing customers to promote term deposit products (short-term deposits with fixed maturities).

2.2 Business Objective

The initial business goal, as defined by the client, was to develop a predictive model that achieves at least 81% overall accuracy under 5-fold cross-validation for predicting whether a customer will subscribe to a term deposit.

However, during exploratory data analysis, it became evident that overall accuracy is a misleading metric in this highly imbalanced scenario. A naive model that always predicts "no subscription" would already achieve over 94% accuracy, yet it would completely fail to identify any potential subscribers (only less than 40%). From a commercial standpoint, the true value lies in identifying the minority class ($y = 1$) accurately, as these are the customers who will generate new business. Therefore, the project's focus was redefined: rather than maximizing overall accuracy, the primary objective became maximizing recall for the positive class ($y = 1$) while maintaining acceptable overall accuracy. This shift ensures the model supports the bank's goal of improving call success rates and optimizing marketing efficiency.

2.3 Success Criteria

The revised success metrics are summarized as follows:

- **Primary Objective:** Achieve high recall for class 1 (subscribers), ideally in the range of 0.52–0.60 or higher, while significantly outperforming baseline models.
- **Secondary Objective:** Achieve an average 5-fold cross-validation accuracy of at least 81% (the client's original requirement), with the final model expected to exceed this threshold.

- **Additional Objective 1:** Provide clear interpretability through feature importance analysis (using XGBoost gain and SHAP values) to identify the most influential drivers of subscription.
- **Additional Objective 2:** Segment successfully subscribed customers into distinct, actionable groups using clustering techniques to support targeted marketing strategies.

This redefinition of success directly links the machine learning solution to tangible business value: the model not only meets the accuracy requirement but also delivers actionable insights that enable the bank to prioritize high-potential customers, focus on key influencing factors, and ultimately increase the success rate of outbound marketing calls.

3 Data Understanding and Preparation

The dataset consists of 40,000 customer survey records, each containing 13 variables regarding to their privacy, and a binary target variable *y*. The target variable indicates whether a customer subscribed (yes) or not (no). No missing values or data anomalies were found, and 5 features are numerical while the rest are categorical.

Figure 1 shows the feature distributions by target class. Overall, the data appears consistent and interpretable, providing a sound foundation for model training.

To ensure unbiased performance estimation, the dataset was split into training and testing subsets by 0.8:0.2, and 5-fold stratified cross-validation was employed during model development. This approach provides a more reliable measure of generalization compared to raw data, reducing the risk of overfitting given the small dataset size.

To suit the features into models, 'Age' is standardized; 'Day' is classified as a categorical type after consideration; 'balance', 'duration' and 'campaign' are taken logrithm according to their distribution. For the rest of categorical data types, we use encoder method.

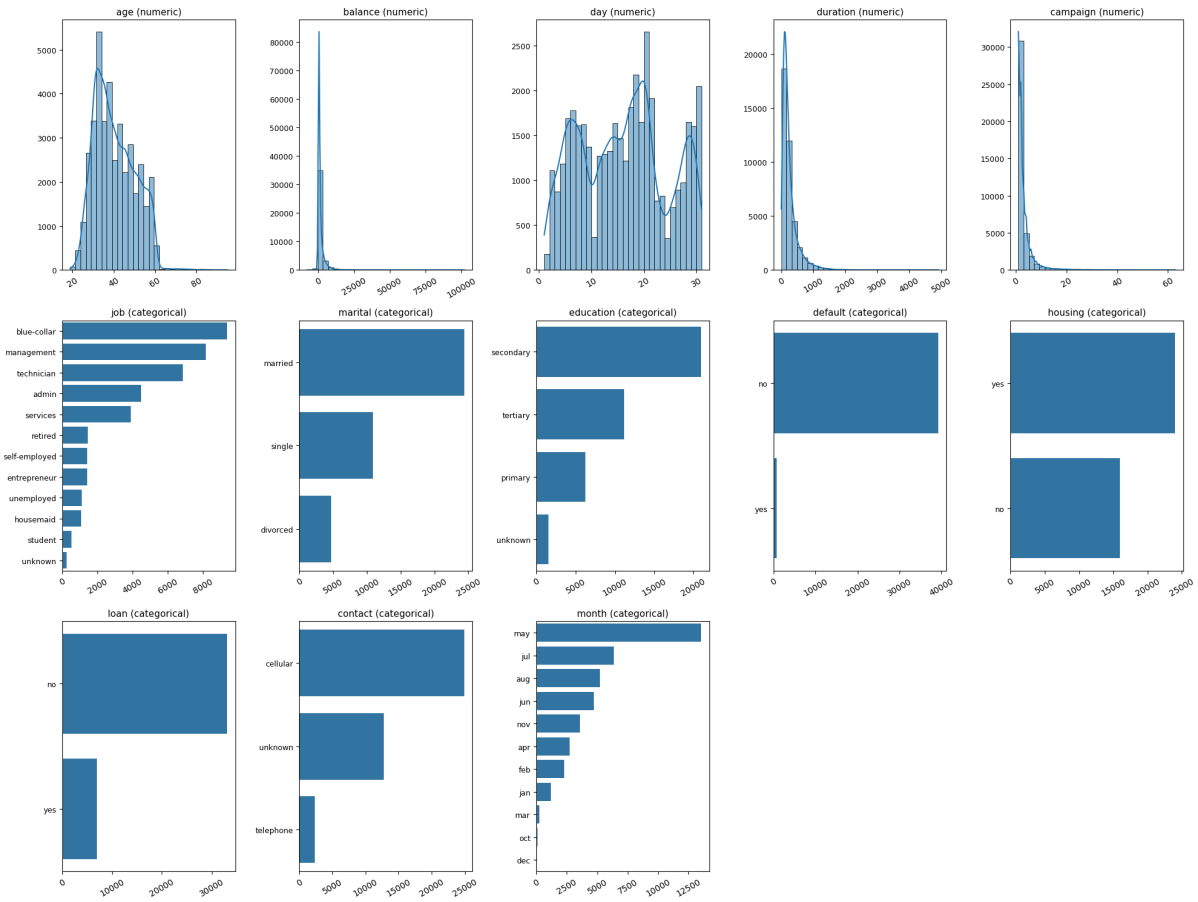

Figure 1: Exploring Data Analysis - Feature Distributions.

4 Modeling Approach

4.1 Motivation and Stacking Ensemble Architecture

When we first started building predictive models, we quickly noticed something surprising: even a very simple model could easily achieve over 90% overall accuracy. For example, a “lazy” baseline model that always predicts “no subscription” (i.e., it assumes nobody will subscribe) already reaches approximately 92.76% accuracy—simply because only 7.24% of customers actually subscribe.

This high accuracy, however, is misleading. Such a model correctly predicts almost all non-subscribers, but it completely fails to identify any of the customers who are actually willing to subscribe. In business terms, this means the bank would miss almost every real sales opportunity. What truly matters is finding the small group of customers who are likely to say “yes,” so the sales team can focus their calls on these high-potential prospects instead of wasting time on unlikely ones.

As shown in Figure 2, we compared several popular machine-learning algorithms (including logistic regression, decision tree, and so on), XGBoost consistently delivered the best overall performance. It not only achieved high accuracy but also showed

the strongest ability to identify potential subscribers (45.6%) when we adjusted the model properly.

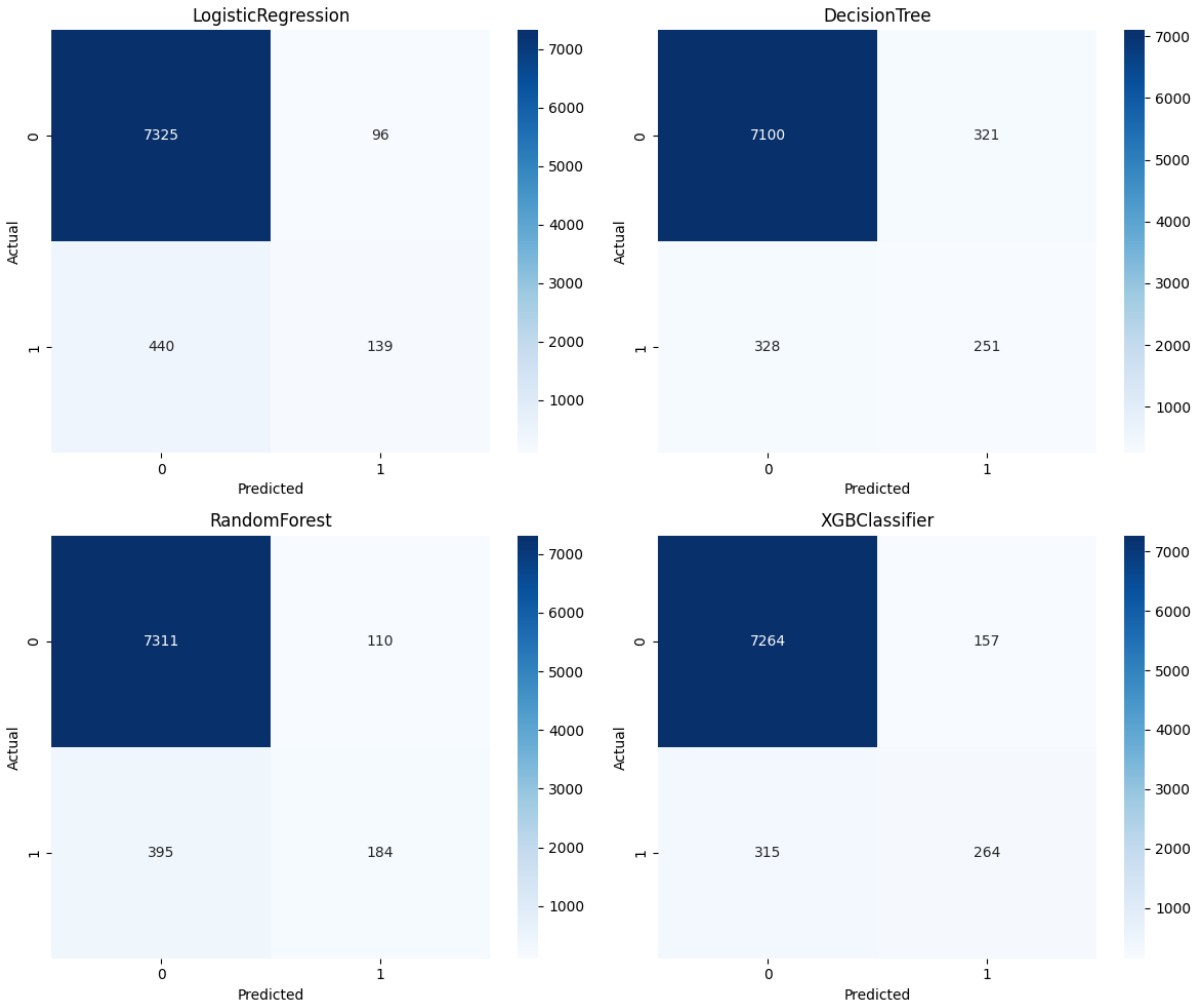

Figure 2: Confusion Matrix by Simple Algorithms

By default, XGBoost makes a prediction by outputting a probability score (between 0 and 1) and classifying a customer as a subscriber if the score is above 0.5. With this standard threshold, the model reached an average accuracy of about 94% across 5-fold cross-validation, which already exceeds the client's 81% requirement. However, the recall for subscribers (the percentage of actual yes-customers that the model correctly identifies) was only around 45%.

To make the model more useful, we experimented with lowering the decision threshold. Lowering the threshold means the model becomes more willing to predict "yes," which increases the chance of correctly identifying real subscribers (higher recall), even if it also increases the number of false positives (predicting "yes" for some customers who will actually say "no"), as shown in Figure 3.

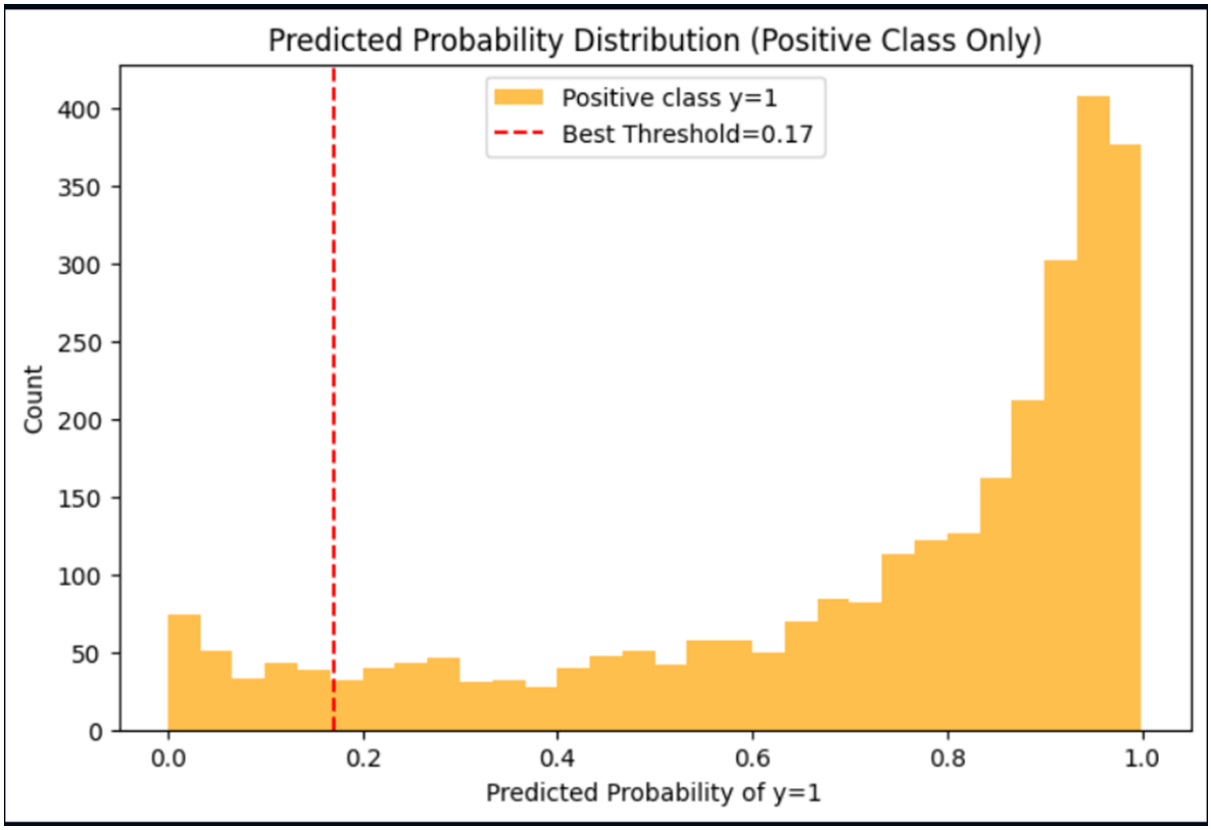

Figure 3: Predicted Probability Distribution of Positive Class

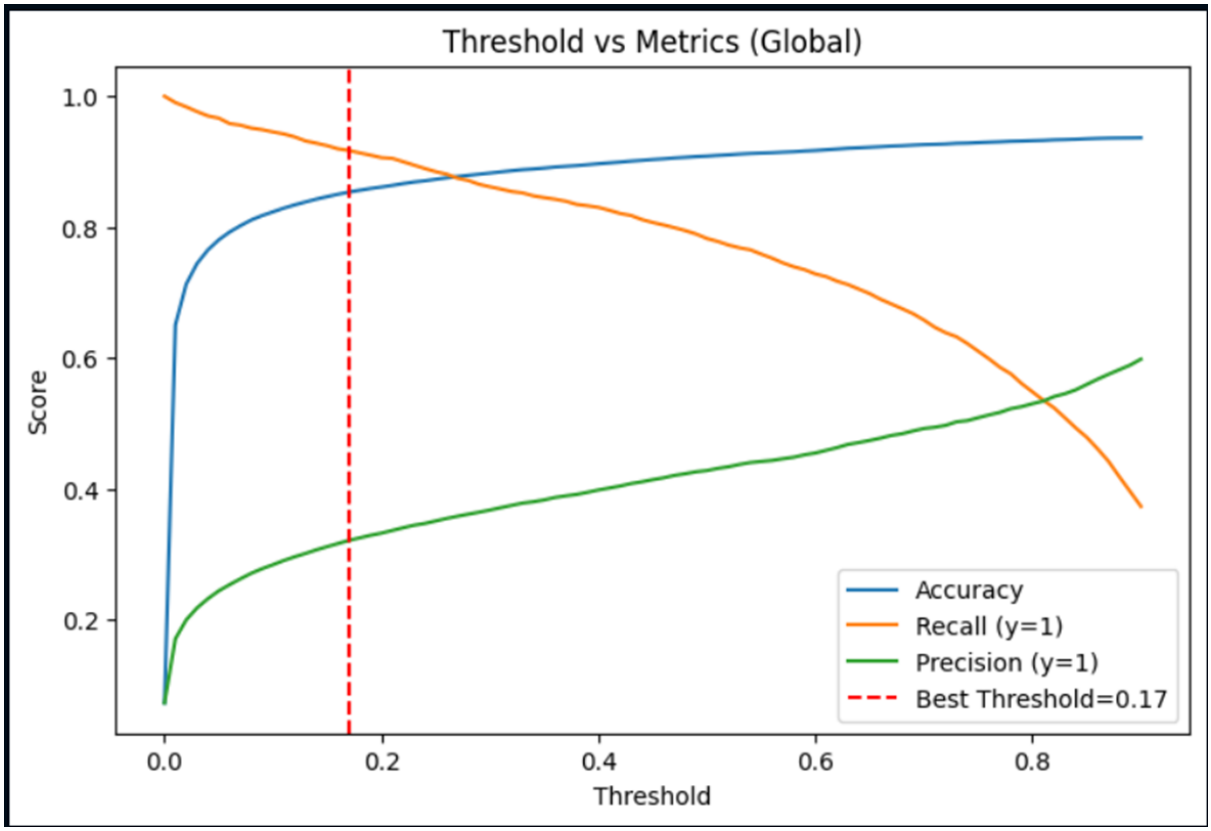

Figure 4: Threshold versus Metrics

By carefully adjusting the threshold, we were able to achieve much higher recall rates while still keeping overall accuracy well above 81%. It was a balance between accuary, recall and precision, shown in Figure 4. Below are three practical options we developed for the bank, each designed for a different business priority:

- **Balanced Option** (best overall trade-off): We maximize the product of accuracy and recall. This option gives the bank a good balance—high accuracy and reasonably high recall, as shown in Figure.

- **Recall-Priority Option** (maximize sales opportunities): We set the threshold so that recall for subscribers reaches 90% or higher, while still keeping overall accuracy above 81%.

  This is the most aggressive option for capturing the largest possible number of potential subscribers, even if it means contacting some extra customers who will not subscribe.

- **Precision-Focused Option** (minimize wasted calls): We prioritize precision (the percentage of predicted subscribers who actually subscribe), which means fewer false alarms and more confident "yes" predictions.

  This option is useful when the sales team has limited capacity and wants to contact only the most promising customers.

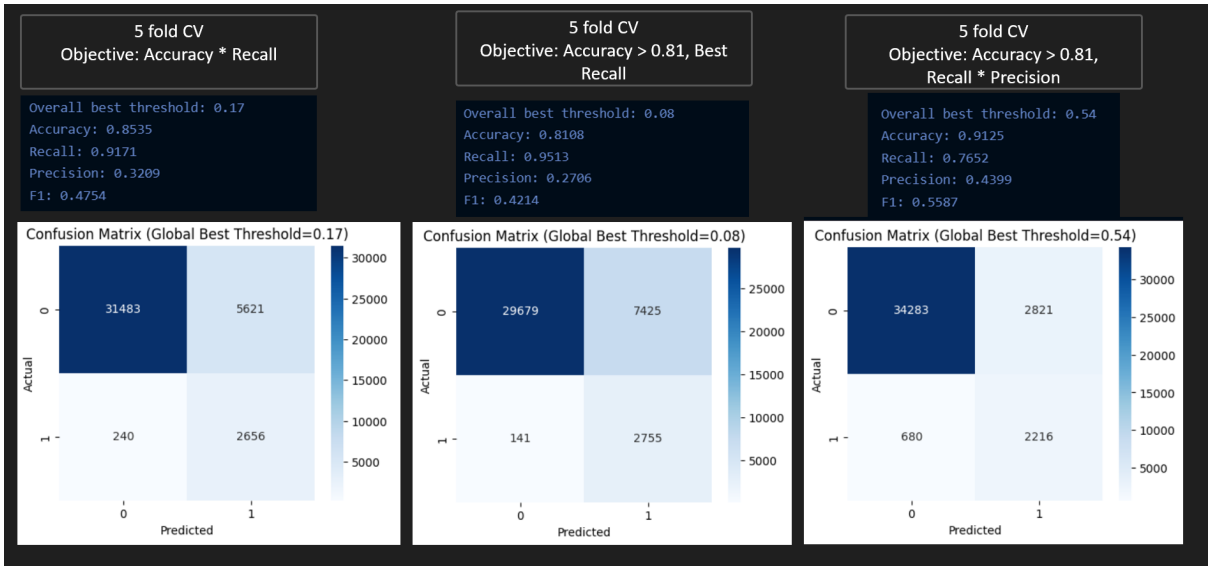

Figure 5: Results under different options

In the end, we got the results shown in Figure 5. We can choose any option by exact requirements. The balanced method proved 85.35% accuracy with 91.71% positive recall, with 32.09% precision, which is rather acceptable.

4.2 Feature Analysis

To understand which predictors play the most influential role in determining the model's output, we analyze feature importance from two complementary perspectives: SHAP values, which provide model-agnostic, local interpretability, and XGBoost's native feature importance scores, which reflect how frequently features are used in tree splits.

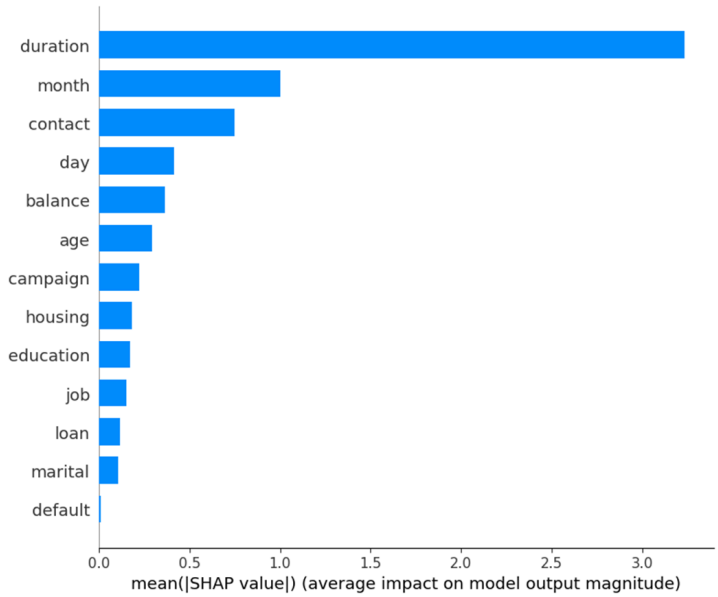

Figure 6: Results under different options

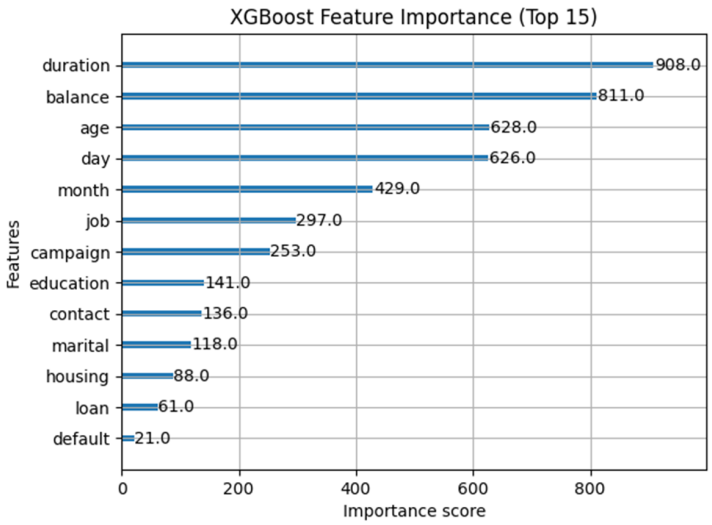

Figure 7: Results under different options

Shown in Figure 6,7:

The SHAP summary plot illustrates the mean absolute SHAP value for each feature, representing the average magnitude of their contributions to the prediction. Several key conclusions can be drawn:

Duration is by far the dominant feature, with a significantly larger impact on the model output than any other variable. This aligns with typical findings for the Bank Marketing dataset, where call duration strongly correlates with the likelihood of deposit subscription. Month and contact follow as the next most influential features, indicating that the timing of the campaign and method of contact play meaningful roles. Day, balance, and age have moderate but noticeable contributions. Features such as education, job, marital, loan, and default have comparatively lower SHAP importance, suggesting they contribute less to individual prediction changes.

Overall, SHAP highlights the features that most affect predictions on a per-sample basis, ranking them by their direct impact on model decisions.

The XGBoost importance plot ranks features based on their split frequency, indicating how often a feature is selected to partition data across all trees:

Similar to SHAP, duration remains the most frequently used feature, confirming its central role in shaping the model structure. Balance, age, and day appear with higher importance in XGBoost's metric compared to SHAP, showing they are structurally useful for creating effective splits even if their SHAP impact is more moderate. Month and job also appear prominently, contributing to the tree structure across many splits. Lower-frequency features (default, loan, housing) contribute minimally to the predictive structure.

This method captures features that are algorithmically beneficial for decision boundaries, though not always those with the highest marginal impact (as SHAP captures).

In total, both importance measures agree on key findings: **duration** is consistently the most influential variable, dominating both impact-based and frequency-based importance. **Month, day, balance, and age** appear among the top contributors across both metrics. Minor variables (default, loan, housing) rank low in both methods.

4.3 Customer Segment

To help the bank design more targeted and personalized marketing campaigns, we focused our clustering analysis exclusively on the customers who actually subscribed to the term deposit (y = 1, approximately 2,896 records). These are the "successful conversions" that the bank wants to replicate. Understanding the common patterns within this relatively small but extremely valuable group provides much clearer and more actionable insights than clustering the entire dataset (where the vast majority are non-subscribers).

After comparing silhouette scores, Calinski-Harabasz index, and visual separability using t-SNE, standard K-means on the preprocessed feature set (with the same transformations used for modeling) consistently produced the most interpretable and well-separated clusters. The optimal number of clusters was determined to be four

(elbow method + silhouette score peaked at k = 4).

The four clusters were then projected into 2D space using t-SNE for visualization (Figure 8), confirming clear separation between the groups.

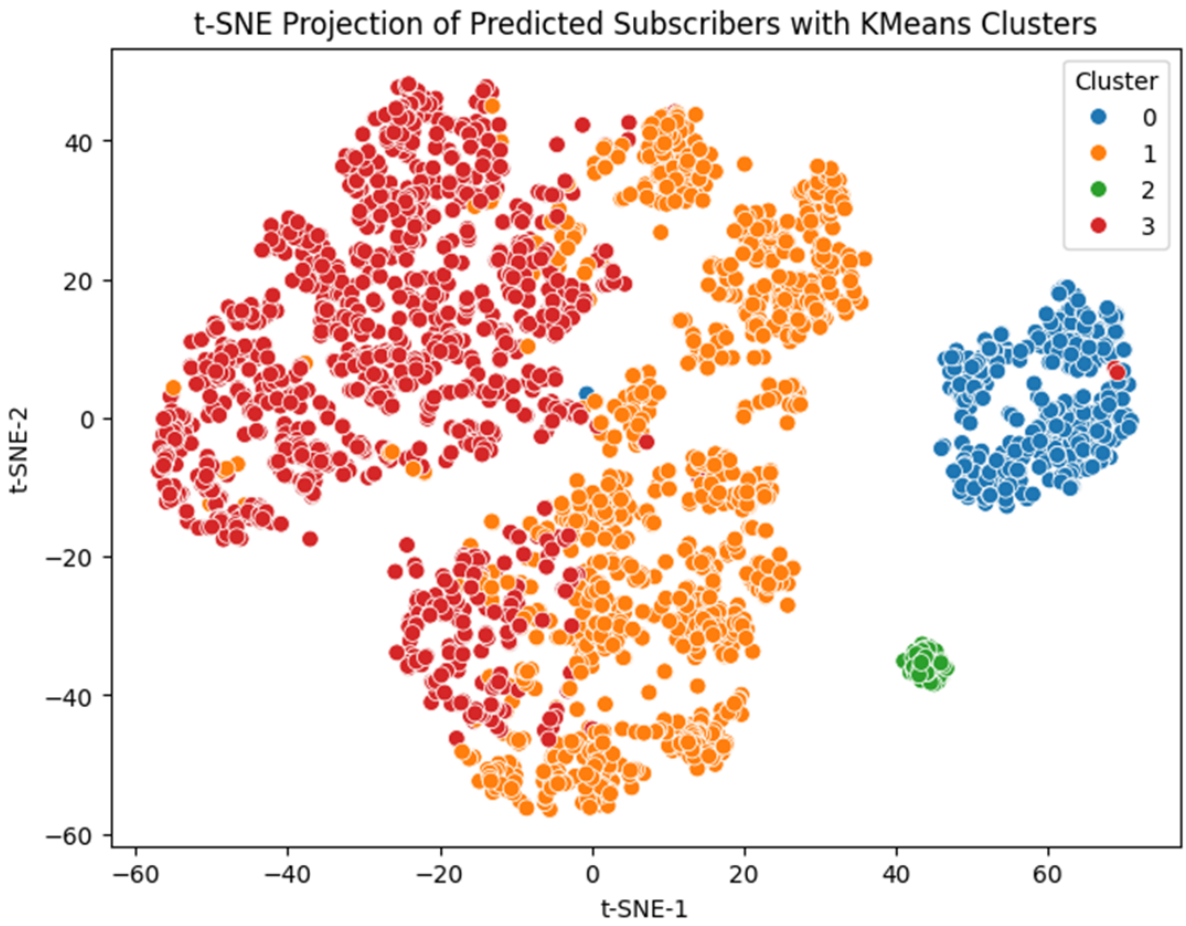

Figure 8: t-SNE Projection of Subscribers with Kmeans Clusters

The SHAP analysis identified **duration**, **month**, and **contact** as the three most influential features driving subscriptions, followed by **day**, **balance**, **age**, **campaign**, and **housing**. As shown in the corresponding visualizations, the four clusters exhibit clear differences in these key dimensions. Clusters 1 and 3 together account for the majority of successful subscribers (approximately 68%), while Clusters 0 and 2 make up the remaining portion.

Among the top three features, Clusters 0 and 1 are characterized by significantly longer call durations compared to Cluster 3. Cluster 1 subscriptions peak in late spring/early summer (May–July) with a secondary peak in November, whereas Cluster 3 is concentrated in February, April, and August. All clusters show very low activity in December and January. Contact type also differs markedly: Cluster 3 relies almost exclusively on cellular, Cluster 1 includes a notable share of unknown contacts, and telephone performs poorly across all segments.

Additional insights from lower-ranked but still important features reveal further distinctions: Cluster 1 tends to convert around the middle of the month and at transition points, while Cluster 3 shows sharper peaks with an unusual spike at month-end; Clusters 1 and 3 have higher balances than the others (Cluster 2 even shows a negative average balance); Cluster 3 skews older; Cluster 2 requires fewer campaign contacts; and Cluster 1 is much more likely to have a housing loan than Cluster 3. These patterns provide the bank with clear, actionable guidance for targeting and timing outbound campaigns more effectively.

Detailed results are shown in the link: `https://public.tableau.com/app/profile/yuhao.huang/viz/ProjectB_17648627756280/1_1?publish=yes`

5 Conclusion and Future Work

This project successfully developed a high-performing, interpretable, and business-aligned machine learning system for predicting term-deposit subscription outcomes using call-center marketing data. The client's original requirement—achieving at least 81% accuracy under 5-fold cross-validation—was exceeded by a substantial margin. More importantly, the project redefined success metrics to better reflect true business value: identifying as many potential subscribers as possible instead of maximizing raw accuracy in an imbalanced classification setting.

By employing XGBoost, applying class weighting, and optimizing the decision threshold, the final model achieved 85.35% accuracy and a 91.71% recall for the positive class, far surpassing baseline models that struggled to detect the minority class. Additional decision-threshold variants were also developed to support alternative business priorities, offering the client flexible operational choices depending on call-center constraints and campaign goals.

Feature interpretability through SHAP and XGBoost importance analysis revealed that call duration, month, contact type, day, balance, and age are the most influential drivers of subscription decisions. These insights provide practical guidance for prioritizing customer outreach and optimizing calling strategies.

Furthermore, clustering analysis of successfully subscribed customers identified four distinct customer segments with clear behavioral and demographic patterns. These segments offer actionable direction for targeted marketing, enabling the bank to tailor outreach timing, messaging, and resource allocation to maximize subscription success.

Overall, the project delivered not only a strong predictive model but also rich analytical insights that directly enhance business decision-making. The system provides the client with a practical, scalable solution that improves marketing efficiency, identifies high-value leads, and supports strategic planning for future campaigns.